

Risk Analysis of Pension Portfolio Liabilities in the Netherlands

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Abstract

This report provides valuation of the pension plan liabilities for ISSurance, on the Green, Silver and Gold portfolios within the Netherlands. The analysis begins by exploring the historical evolution of mortality across both time and age. It is followed by the calibration of Lee-Carter model to analyze the a_x , b_x , and k_t parameters. To take lifespans exceeding standard data limits into account, the Kannisto method is applied to extrapolate mortality rates and determine the period life expectancy and Expected Present Value (EPV) of the liabilities. Then, the scope is expanded from period to cohort thinking. This reveals differences in valuation when future mortality improvements are modeled. Later, a holistic risk assessment is achieved through the simulation of mortality paths and the application of Solvency II stress tests for mortality and financial shocks. These simulations allow for a deeper assessment of the uncertainty surrounding the best estimate of total liabilities. Finally, the report concludes by projecting the total portfolio cashflows over time. This provides insights on the liquidity required to cover liabilities under various scenarios including stress scenarios.

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1 Historical Evolution of Mortality in the Netherlands

1.1 Data and its Structure

To build the valuation of pension liabilities, it is essential to first understand the underlying mortality dynamics of population. To achieve this, we utilize the Lee-Carter model [1] as it provides a framework for forecasting future mortality trends and generating realistic future scenarios. The model is specified as:

$$\ln(m_{x,t}) = a_x + b_x k_t + \epsilon_{x,t} \quad (1)$$

In this specification, a_x represents the average mortality level at age x , and k_t tracks the general mortality trend over time. The b_x parameter measures how sensitive mortality at age x is to changes in the overall trend while $\epsilon_{x,t}$ accounts for residual variation.

Because the central death rate ($m_{x,t}$) is the fundamental input required to calibrate the Lee-Carter model, we have sourced historical mortality data for the Netherlands from the Human Mortality Database (HMD) [2]. Specifically, I utilize 1×1 tables, representing one-year age intervals and one-year time intervals, for both death counts ($D_{x,t}$) and exposure-to-risk ($E_{x,t}$).

Inspection of the raw data shows records from the year 1850 up to 2023, covering ages from 0 to 110+.

```
> head(nld_deaths)
  Year Age Female Male Total
1 1850  0   8735 10804 19539
2 1850  1   2292  2454  4746
3 1850  2   1009  1055  2064
4 1850  3    591   634  1225
5 1850  4    461   521   982
6 1850  5    411   425   836
...
> tail(nld_deaths)
  Year Age Female Male Total
19309 2023  105  34.30  4.29 38.59
19310 2023  106  18.77  2.18 20.95
19311 2023  107   9.46  1.00 10.46
19312 2023  108   4.72  0.27  4.99
19313 2023  109   2.30  0.00  2.30
19314 2023 110+   1.32  0.00  1.32
```

1.2 Calculation of Central Death Rates

The raw counts of deaths ($D_{x,t}$) and exposures ($E_{x,t}$) must be converted into central death rates to calibrate the Lee-Carter model. The central death rate represents the average risk of death for an individual aged x , during the calendar year t . It is defined as the ratio of total deaths to the total exposure to risk, in this case exposure is the person-years lived.

$$m_{x,t} = \frac{D_{x,t}}{E_{x,t}} \quad (2)$$

As $m_{x,t}$ is not a standard probability, it can take greater values than 1. Because the denominator represents person-years rather than the initial number of individuals, the central death rate can exceed 1. For example, consider a group of 10 individuals observed at the start of the year. If all 10 pass away within the first month, they collectively contribute only $10 \times \frac{1}{12} \approx 0.833$ person-years of exposure. So, the central death rate becomes $m_{x,t} = 10/0.833 = 12$.

To compute our central death rates, we join the deaths and exposures datasets. While doing that, we filter out the "110+" age category at this stage as the variable x is vague in this category. Also, removing this interval ensures that the age variable is numeric.

1.3 Visualizing Historical Trends

To explore the evolution of mortality across different generations in our portfolio, we isolate specific decade ages (30 through 90) and plot the natural logarithm of their central death rates over time. The resulting visualization is presented in Figure 1 below.

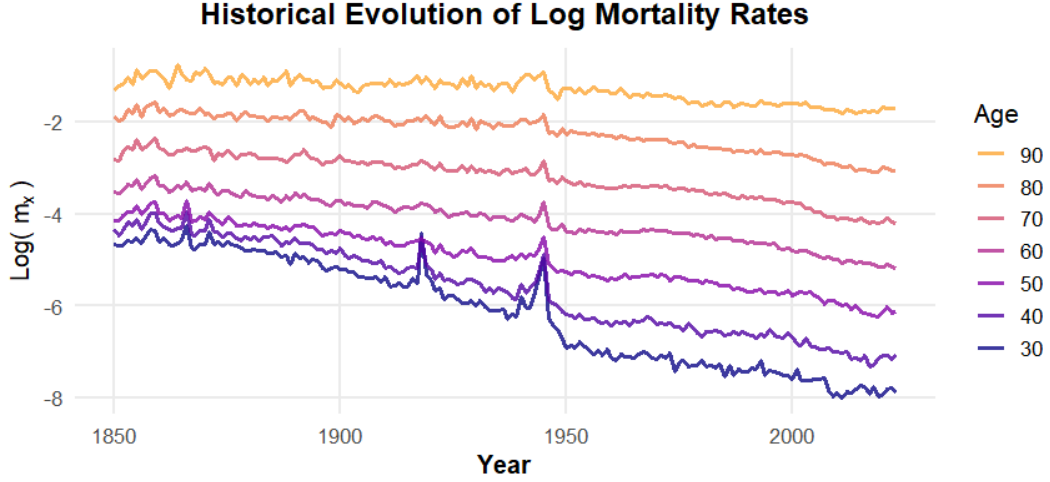


Figure 1: Historical evolution of log-mortality rates in the Netherlands (1850–2023) across ages.

Commentary on the Evolution of Mortality:

Observing the plotted $\log(m_{x,t})$ curves, several demographic trends do appear:

- **General Downward Trend:** There is a clear long-term downward trajectory in mortality rates across all age groups. This shows historical improvements in healthcare and the living standards in the Netherlands.
- **Age Stratification:** The lines are more or less parallel to each other and stacked chronologically by age. As expected, mortality tend to increase with age. So, a 90 year old person has a higher risk of death than a 80 year old person regardless of the year.
- **Historical Shocks:** The plots capture some shocks, most notably the spikes corresponding to the 1918 Spanish Flu pandemic and the periods around the World Wars. An increase in recent years is also visible which shows the impact of the COVID-19 pandemic.
- **Varying Rates of Improvement:** The slope of improvement is not identical across ages. Younger ages show steeper historical declines in mortality compared to older ages. This indicates that medical advancements have had a large relative impact on preventing early-to-mid-life deaths.

1.4 Fitting the Lee-Carter Model

To calibrate the Lee-Carter model, the raw data is reshaped into a matrix format (years by ages). For the initial calibration phase, we restrict the age range to 0 – 90. Mortality rates for old ages (90+) show volatile movements possibly due to low exposure. This will be addressed using the Kannisto extrapolation method later. For now, the following implementation prepares the death ($D_{x,t}$) and exposure ($E_{x,t}$) matrices and fits the standard Lee-Carter model:

```

> Dxt[1:5, 1:6]
  1850  1851  1852  1853  1854  1855
0 19539 21058 24000 21248 20749 22228
1  4746  5549  5825  5138  5479  6913
2  2064  2723  2557  2582  2820  3863
3  1225  1317  1394  1497  1674  2364
4   982   926   805   986  1060  1663
...
> Ext[87:91, 170:174]
      2019      2020      2021      2022      2023
86 56446.92 57311.46 59116.00 59981.57 61718.86
87 50798.45 50394.30 51000.94 52893.15 53866.87
88 44520.35 44662.15 44052.29 44807.17 46743.62
89 37206.62 38472.43 38378.07 37917.50 38702.46
90 30747.90 31498.30 32395.15 32395.21 31970.28

```

Once the model is fitted, it decomposes the mortality into three distinct parameters: a_x , b_x and k_t . To present these findings, we extract the parameters from the fitted object and visualize them in Figures 2, 3, 4.

Comments on the Lee-Carter Parameters:

The calibration gives insights into the demographic profile of the Netherlands:

- **Baseline Profile (a_x):** This parameter represents the average log-mortality across all years. As expected, it demonstrates a tick shape: First, high infant mortality, then dropping to a minimum during around 10-15 and stabilizing around twenties and finally rising linearly with older age.
- **Improvement Sensitivity (b_x):** This indicates which age groups benefit most when the general mortality trend improves. The higher values observed at younger ages (infants and early adults) confirm that medical advancements have mostly reduced premature deaths. On the other hand, they have less relative impact on the oldest age groups, confirming our observations in Figure 1.
- **Mortality Trend (k_t):** This time dependent index tracks the overall direction of mortality throughout the years. The downward trajectory confirms consistent historical improvements in life expectancy excluding the shocks mentioned above.

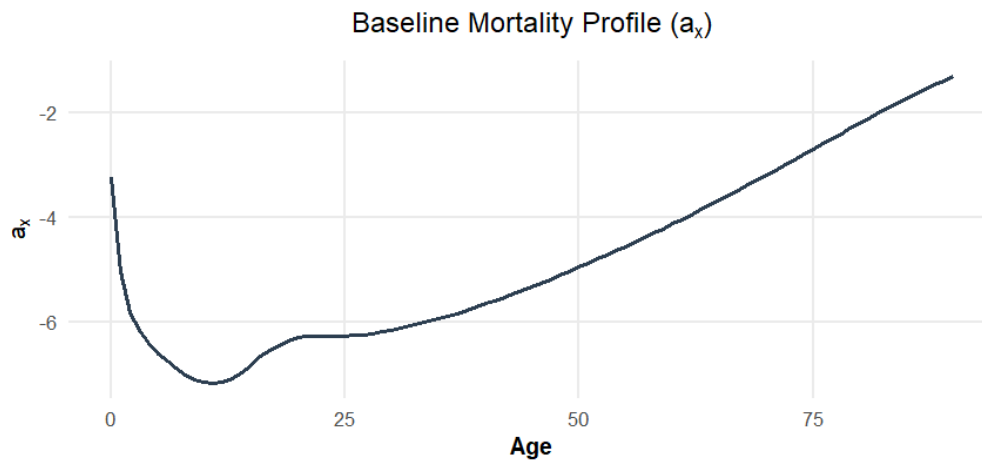


Figure 2: Baseline Profile (a_x)

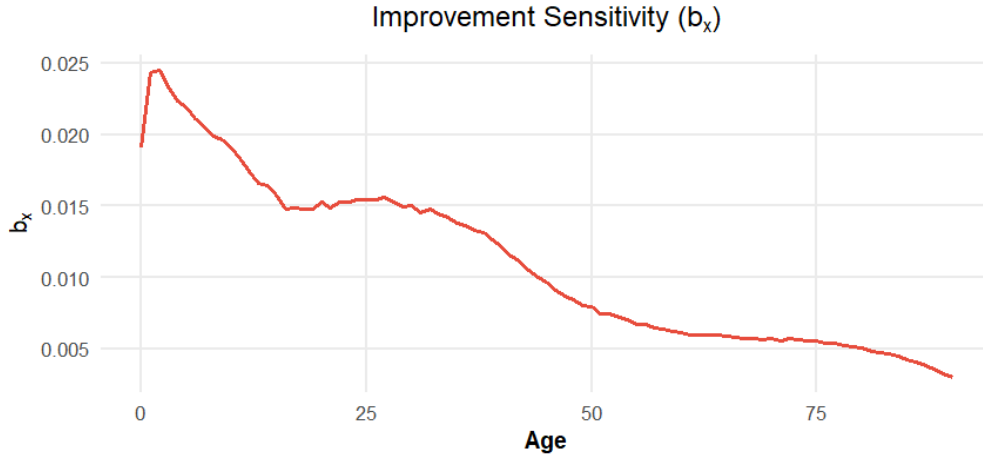


Figure 3: Improvement Sensitivity (b_x)

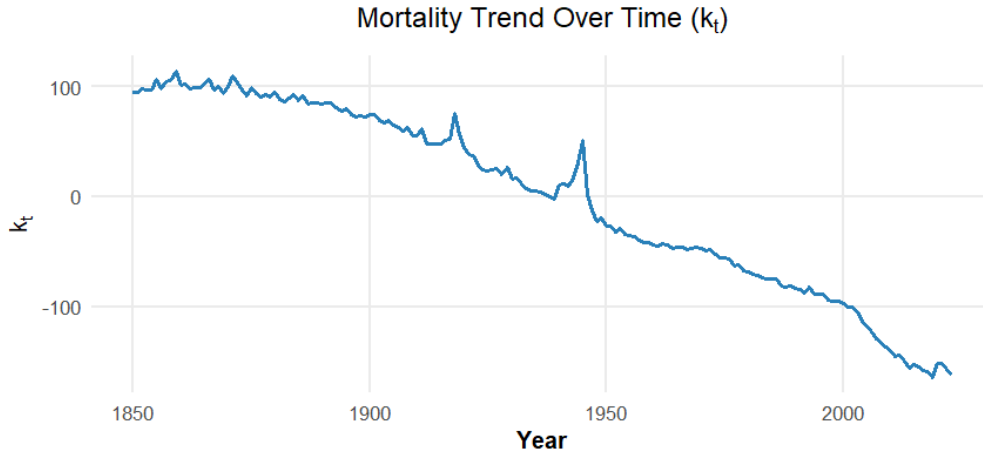


Figure 4: Mortality Trend (k_t)

1.5 Old-Age Mortality: The Kannisto Method

An important parameter in valuing pension liabilities is the ultimate age of the life. Defining the ultimate age is important to capture the total expected duration of payments. For this portfolio, the youngest participants are currently 30 years old. To ensure the complete liabilities over a 90-year projection horizon, the ultimate age is set to 120.

However, historical mortality data of the Netherlands at older ages, for example 90+, is sparse for direct calibration. To illustrate this sparsity, we can observe the raw HMD death counts ($D_{x,t}$) from the year 2023. While middle-ages have relatively high sample sizes, data reduces exponentially at advanced ages:

```
nld_data %>% filter(Year==2023, Age %in% c(60,105,108))%>% select(Age,Total.x,Total.y)
```

	Age	Total.x	Total.y
1	60	1396.00	249113.28
2	105	38.59	57.22
3	108	4.99	5.28

With only 5.28 person-years of exposure at age 108, calculating empirical central death rates becomes noisy. To resolve this and make the completion of the mortality table smoother, we apply Kannisto

method [3]. The Kannisto method is based on the assumption that the logit transformation of the central death rate follows a linear relationship with age.

The procedure is implemented as follows:

1. **Estimation:** We isolate the fitted Lee-Carter mortality rates for ages 80 to 90. Then, OLS regression is fitted to the logit of these rates:

$$\text{logit}(m_{x,t}) = \ln \left(\frac{m_{x,t}}{1 - m_{x,t}} \right) = \phi_1 + \phi_2 x \quad (3)$$

It is important to note that the logit function is mathematically defined only for inputs strictly between 0 and 1. If any raw central death rate equals or exceeds 1, the transformation fails. Because of this, we fit the regression using the stabilized rates from ages 80 to 90 rather than the volatile raw data at age 105+.

2. **Extrapolation:** Using the estimated coefficients ϕ_1 and ϕ_2 , the linear relationship is projected forward for ages 91 to 120.
3. **Transformation:** The inverse logit function is applied to the extrapolated values to return them to standard central death rates. Because the inverse logit function mathematically asymptotes at 1, it forces the extrapolated rates to be strictly bounded between 0 and 1. This provides a stable tail for liability valuation but capping $m_{x,t}$ at 1 is a drawback of the model. This might prevent the rate from capturing real-world scenarios where mortality risk exceeds one death per person-year.

To validate this approach, the extrapolated Kannisto tail was plotted alongside the fitted Lee-Carter rates and the raw empirical HMD data (Ground Truth) for the year 2023. As visualized in Figure 5, the Kannisto extrapolation is not bad at capturing the trajectory of real-world mortality at advanced ages.

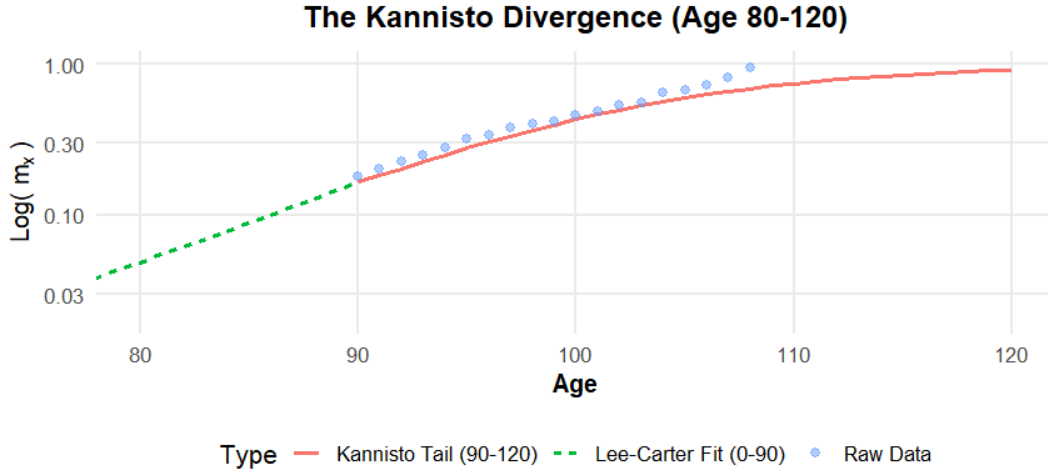


Figure 5: Mortality profile displaying the Lee-Carter fit (ages 80-90) and the Kannisto extrapolated tail (ages 90-120), plotted against the raw HMD ground truth data.

1.6 Evaluation of Period Life Expectancy

With a full matrix of $(m_{x,t})$ up to age 120, now we can evaluate the pension participants life expectancy (e_x) using period approach. In the calculation of period life expectancy, we assume frozen mortality profile. So, for example a 30-year-old today is assumed to experience the exact same mortality risk at age 80 as an 80-year-old experiences today.

The expected remaining lifetime is calculated by integrating the survival probabilities over the remaining lifespan. Assuming a piecewise constant force of mortality between integer ages, the period life expectancy is computed from the projected central death rates. Figure 6 illustrates the historical evolution of period life expectancy for some ages.

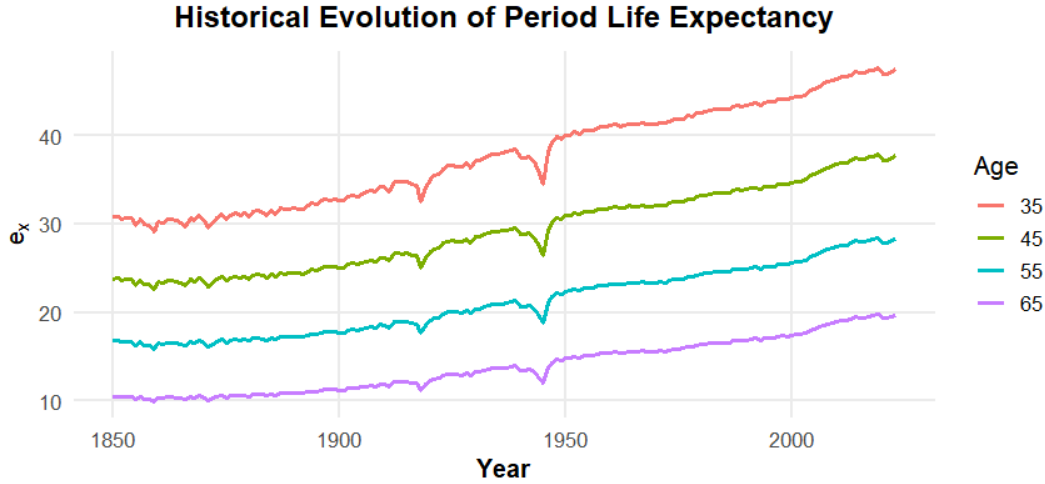


Figure 6: Historical evolution of period life expectancy for selected ages.

The visualization shows a long-term upward trajectory in expected lifetimes across all observed ages. However, just relying on period life expectancy for pension valuation underestimates liabilities. Because it does not account for the future medical improvements that today's 30-year-old will experience as they age. This limitation motivates to use cohort thinking in the next phases of this analysis. Before moving to cohort based life expectancy in the next section, we conclude this section by calculating the expected present value (EPV) of a whole life annuity.

1.7 EIOPA Risk-Free Rate

To accurately value the long-term pension liabilities, future cash flows must be discounted to their present value. For this purpose, we use the risk-free interest rate term structures published by EIOPA.

For this valuation, the baseline discount curve is derived from the EIOPA spot rates for the Netherlands. We used the curve incorporating the Volatility Adjustment (VA). The reason why VA is utilized because pension liabilities are long-term. So, applying the VA prevents artificial fluctuations in the valuation caused by short-term volatility in bond spreads. The extracted term structure is visualized in Figure 7.

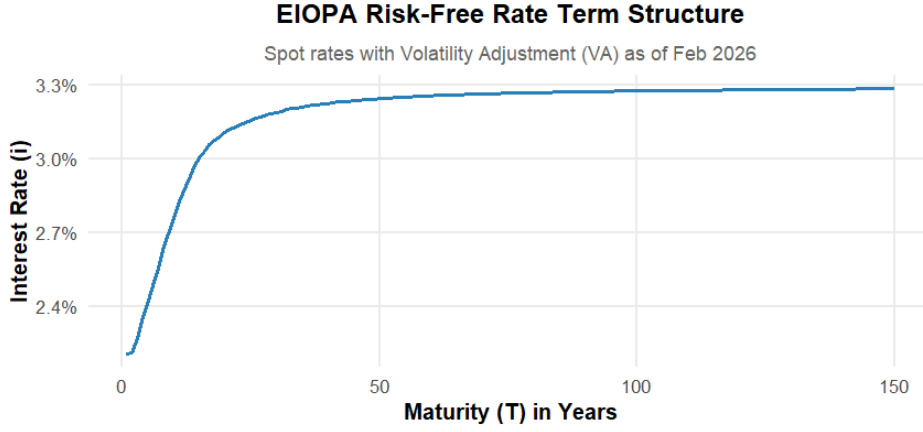


Figure 7: EIOPA Risk-Free Rate term structure for the Netherlands

1.8 Valuation of Period Liabilities

Now, with both the closed mortality matrix and the EIOPA discount curve, we can calculate the Expected Present Value (EPV) of the pension liabilities.

The portfolio consists of lifelong pension benefits payable annually in advance under the condition that the participant's survives. For a participant who has already reached the retirement age ($R = 65$), the EPV of a whole life annuity issued at age x in period t , with an annual payout C is mathematically defined as:

$$\ddot{a}_{x,t} = C \sum_{T=0}^{\infty} v(t, T)^T {}_T p_{x,t} \quad (4)$$

where:

- ${}_T p_{x,t} = \prod_{j=0}^{T-1} p_{x+j,t}$ is the T -year survival probability calculated under the period approach.
- $v(t, T) = \frac{1}{1+i(t, T)}$ is the annual discount factor for maturity T .

However, the ISSurance portfolio includes active participants for example aged 30. For these individuals, the annuity is deferred. To accurately value this liability under the EIOPA risk-free term structure, the valuation adds the discounted cash flows across the participant's remaining lifespan, starting from their current age.

An indicator function, $I(x + T \geq R = 65)$ is applied to make sure that pension payments only start when the retirement age is reached. Each individual future cash flow is then discounted back to the valuation date. So, the Expected Present Value (EPV) of the deferred annuity is mathematically defined as:

$$\text{Deferred EPV}_{x,t} = C \sum_{T=0}^{\infty} I(x + T \geq 65) \times v(t, T)^T \times {}_T p_{x,t} \quad (5)$$

Applying this logic across all portfolio ages where $C = 1$ yields the liability curve shown in Figure 8. For any other payout amount, like 10,000 EUR, the EPV can be multiplied by that amount.

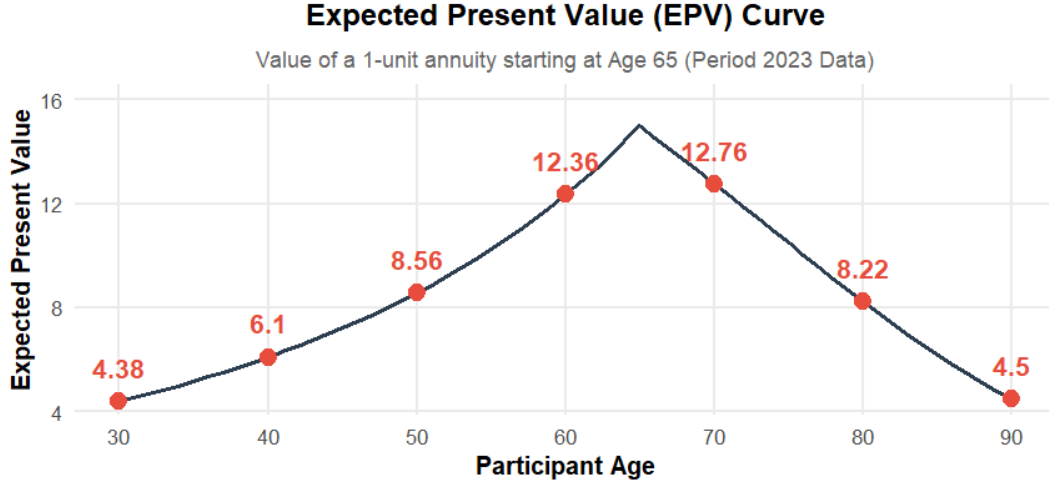


Figure 8: Expected Present Value (EPV) of a 1-unit deferred annuity across portfolio ages.

The curve demonstrates the non-linear relationship between age and expected liability. For younger participants, the EPV is discounted heavily due to the deferral period and the cumulative probability of dying before retirement. As participants approach age 65, the discounting effect diminishes and this drives the EPV to its peak. In the Post-retirement phase, the EPV declines as the expected remaining lifetime shortens. For a rigorous mathematical proof for this observation, see Appendix A.

2 Period vs. Cohort Thinking

The period valuation in previous section relies on a static mortality profile. It assumes that future mortality rates will remain identical to the current year. However, the Lee-Carter trend parameter (k_t) shows that mortality consistently improves over time. To accurately reserve for future obligations, we need to evaluate the valuation under cohort approach.

The difference between these two approaches is best visualized using the diagram below (Figure 9).

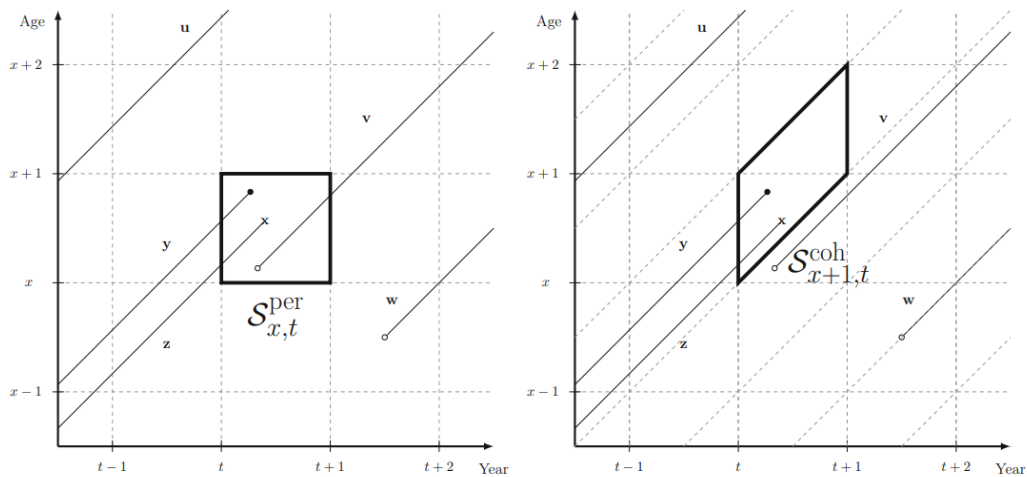


Figure 9: Lexis diagrams illustrating period vs. cohort thinking [4].

The left panel represents period thinking. It takes a vertical slice of data and all mortality rates are extracted from a single, fixed calendar year. For example, it assumes a 50-year-old in 2026 will experience the exact same mortality risk at age 51 as an 51-year-old does today.

On the other hand, the right panel represents cohort thinking. It tracks a diagonal path and as a participant ages by one year, the calendar year also advances by one year. Under this framework, a 50-year-old in 2026 will experience the mortality risk of an 51-year-old in the year 2027. This incorporates expected medical advancements of next year.

2.1 Cohort Life Expectancy

To implement this diagonal tracking, the `LE_period` function was modified into `LE_cohort`. While the period function was extracting mortality rates from a single row corresponding to the valuation year, cohort function extracts a diagonal vector from the matrix: $m_{x+k,t+k}$ for $k = 0, 1, 2, \dots$ until the ultimate age. Because this diagonal extends into the future, the function takes the projected Lee-Carter mortality matrix appended to the historical data as input.

This methodological shift changes the projected survival probabilities. Figure 10 illustrates this divergence by tracking the trajectory of a 30-year-old participant.

During the early years of the projection, the period and cohort curves remain nearly identical because baseline mortality risk during these working years is already low. However, as the participant approaches retirement and advanced old age, a gap emerges. Visually, the solid cohort curve is farther away from the dashed period curve and the drop in survival probability is shifted to the right. Practically, this means a 30-year-old has a higher likelihood of reaching older ages than the period life table approach. This increases the expected duration of pension payouts.

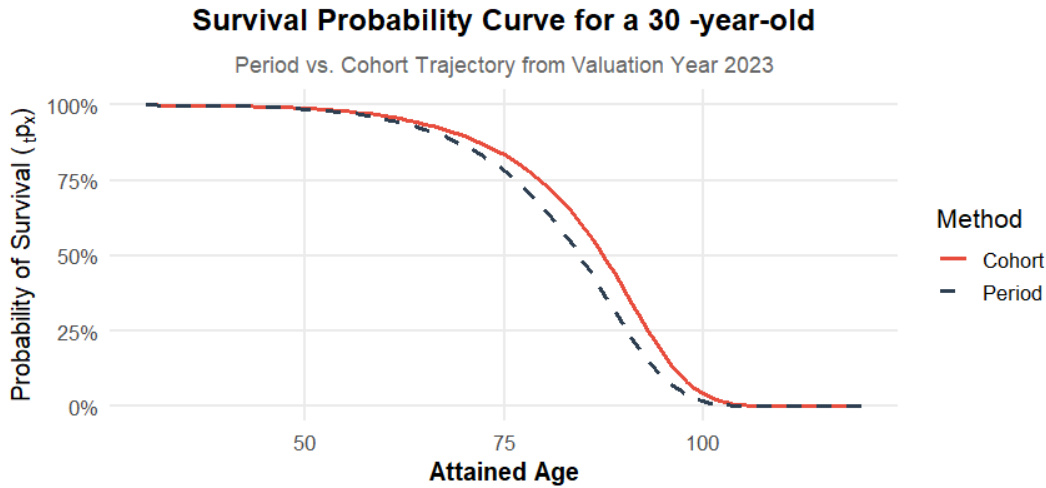


Figure 10: Survival probability curve (${}_tp_{30}$) for a 30-year-old participant.

Though, the impact of this shift depends heavily on age. Figure 11 compares period and cohort life expectancy across the portfolio. The region in between the cohort and the period curves represents the number of years resulting from using cohort approach instead of period approach. Younger participants gain several years of expected life as they have decades of future improvements ahead of them. Conversely, the gap for a 90-year-old is no much as they have a very short horizon for future improvements to show their effect.

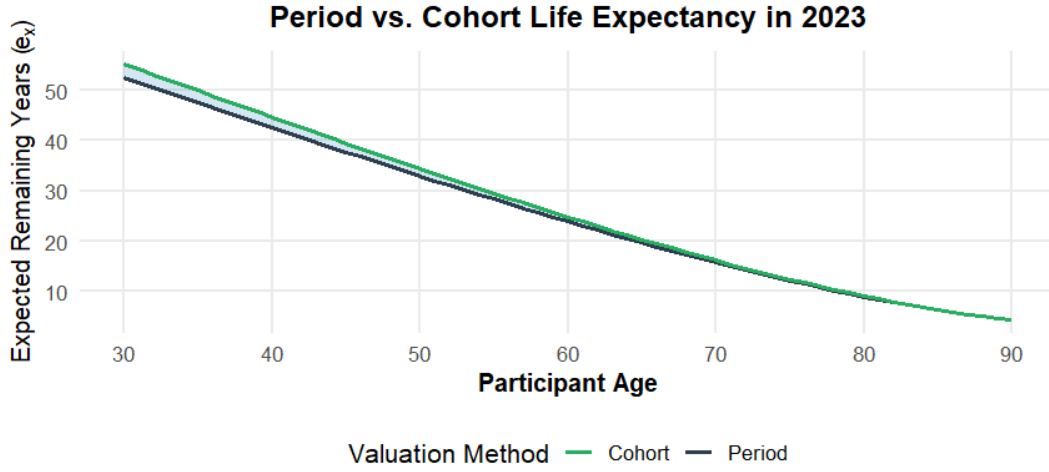


Figure 11: Period vs. Cohort Life Expectancy across portfolio ages.

To show that this divergence is a persistent feature, Figure 12 tracks the expected remaining lifetime of a 65-year-old (e_{65}) from 1970 to 2023. As visualized by the shaded region, the cohort life expectancy is always higher than the period life expectancy across the 50 years of observation period. More, the period line fluctuates corresponding to shocks such as flu seasons or recent pandemic, while the cohort approach gives a smoother path as it integrates these shocks over the participant's remaining lifetime.

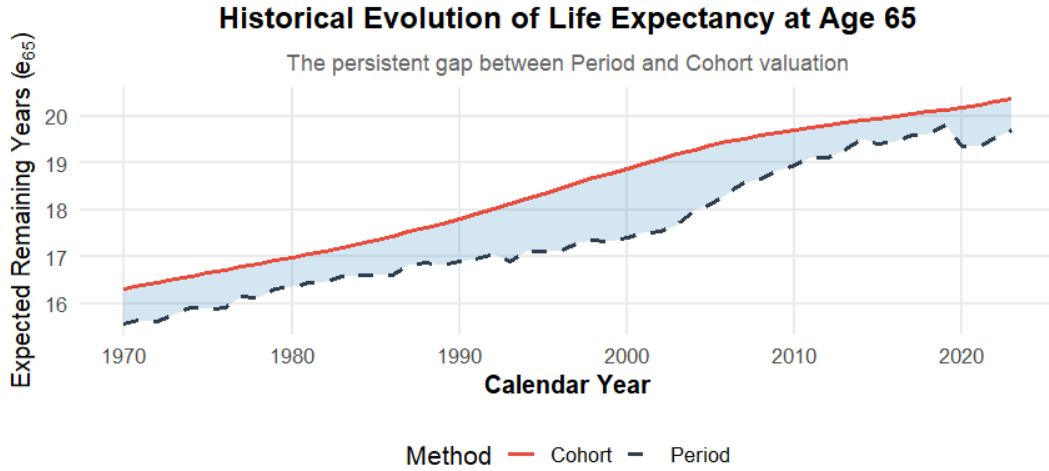


Figure 12: Historical evolution of life expectancy at age 65.

2.2 Valuation of Cohort Liabilities

The divergence between two approaches also impacts the financial valuation of the liabilities. To compute the cohort EPV, we use the same formula above except the mortality input. The cohort valuation changes static period survival probabilities for dynamic, diagonal survival probabilities (${}_t p_x^{\text{coh}}$). In this way, we isolate the financial impact of mortality improvements. Figure 13 maps the resulting EPV difference across ages for a unit annuity.

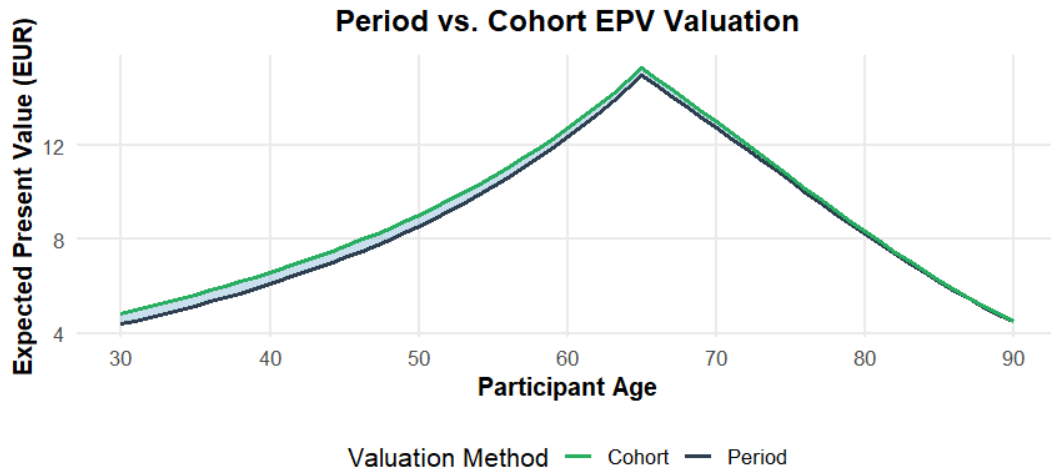


Figure 13: Period vs. Cohort EPV valuation across ages.

The shaded region shows the extra liability resulting from cohort approach. As the expected lifetime increases regardless of the participant age, the number of years that participants will be paid their annuity increases. However, this extra liability differs with the age. For participants with older ages like 80, 90, the two curves converge because their remaining lifespan is too short to experience meaningful medical advancements. To be mathematically rigorous, ${}_tp_x^{\text{coh}} \approx {}_tp_x^{\text{per}}$ when x is large. However, for younger participants (ages 30 to 50), the gap widens. Years of mortality improvements increase their probability of reaching and surviving in their retirement period.

Consequently, a valuation based on static period tables leaves the pension fund under-capitalized. To quantify the financial severity of this shortfall, Figure 14 translates the absolute EPV gap into a relative valuation difference. This metric shows the percentage by which the period method underestimates the true liability over time.

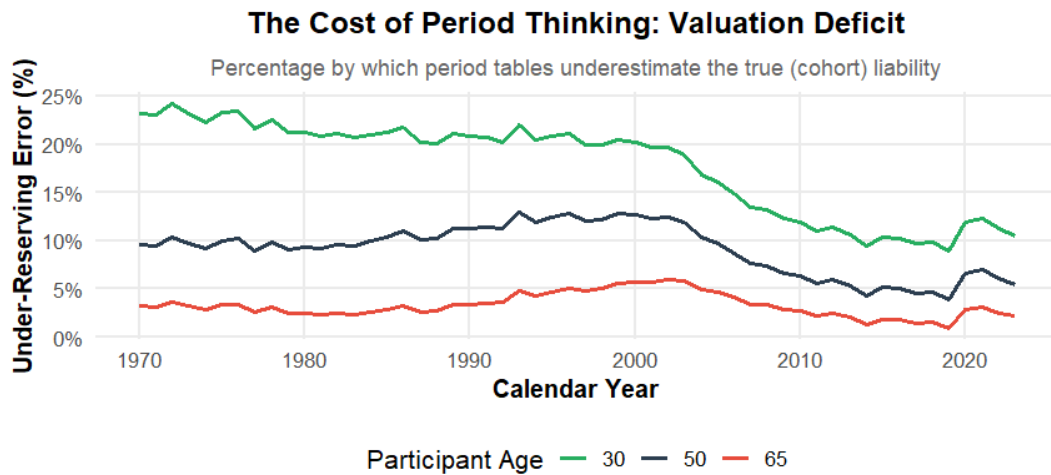


Figure 14: The historical under-reserving error in terms of percentage

This visualization provides an insight regarding age sensitivity. While the period method underestimates the liability for a 65-year-old retiree by a relatively small margin, the error is large for younger active participants. For a 30-year-old in 1970, ignoring future mortality improvements resulted in mispricing the liability by around 25%.

Furthermore, tracking the metric historically shows a downward trend across all age groups. As

baseline mortality rates have been improved from 1970 to 2023, the base probability of survival has increased. So, the relative impact of future improvements on the total EPV has gradually decreased.

Despite this downward trend, the current shortfall remains high. In 2023, period tables still underestimate the liability of a 30-year-old participant around 10%. This sensitivity shows the risk profile of ISSurance's portfolios. As the Green Portfolio is mostly concentrated in younger active members, relying on period tables could create a huge capital shortfall. Conversely, the older Gold Portfolio remains relatively unaffected from this.

2.3 Longevity and Mortality Shocks and their Effects

To capitalize the company sufficiently, insurers better quantify their exposure to demographic scenarios. For a pension portfolio, the primary risks are mortality and longevity shocks. While in the latter, participants live longer than expected and this increases the total payout duration, in the former the risk is that participants die earlier than expected so this results in inefficient use of capital.

To evaluate this uncertainty, we apply two deterministic shocks to the mortality rates ($m_{x,t}$):

- **Longevity Shock:** A permanent 20% decrease in mortality rates ($m_{x,t}$ becomes $m_{x,t} \times 0.80$).
- **Mortality Shock:** A permanent 15% increase in mortality rates ($m_{x,t}$ becomes $m_{x,t} \times 1.15$).

We first evaluate the demographic impact of these shocks. Figure 15 illustrates the life expectancy under the longevity and mortality scenarios.

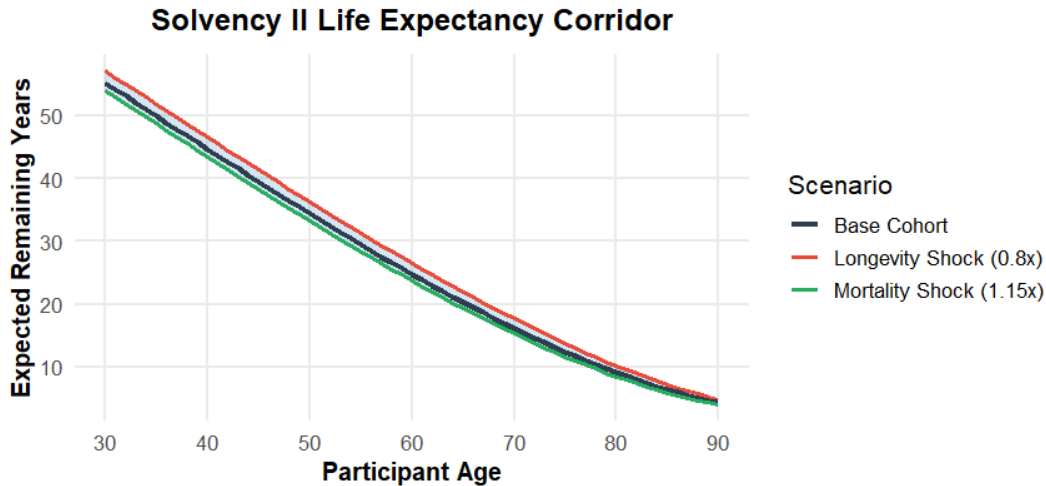


Figure 15: Life Expectancy Risk across portfolio ages.

While the longevity shock (red line) shifts the expected remaining years upward, might be a scenario of sudden medical breakthroughs, the mortality shock (green line) shifts the expectation downward, might be reflecting a public health crisis. Lastly, the shaded region between these scenarios defines the area of biometric uncertainty.

We then translate this demographic uncertainty into financial liabilities. Figure 16 displays the Expected Present Value (EPV) of a unit deferred annuity under the same three scenarios.

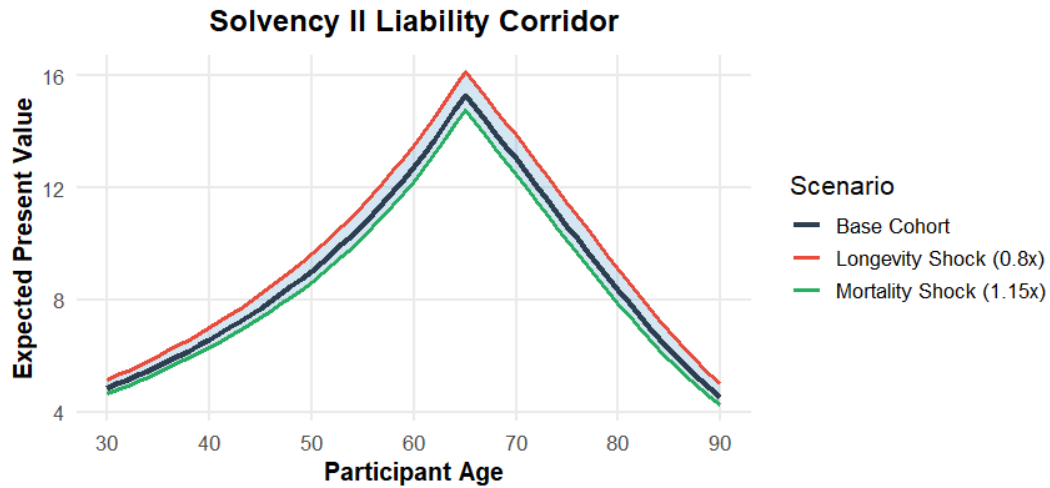


Figure 16: Liability Risk Corridor showing the EPV under baseline and shocked scenarios.

For a pension provider, the longevity shock represents the adverse scenario. While the transition from period to cohort tables impacted younger participants more due to long horizon of projected medical advancements, a static 20% longevity shock impacts older participants more. Because mortality level for a 30-year-old is already extremely low, a 20% reduction yields a negligible increase in their probability of surviving to retirement. However, for participants currently entering or within the payout phase (ages 65 to 75), the baseline mortality risk is much higher. This extends the payout duration when the discount factor is highest. As a result, the capital gap per unit of annuity peaks around age 70s.

```

> df_mort_shocks %>%
  mutate(Abs_Diff = Long - Base) %>%
  arrange(desc(Abs_Diff)) %>%
  select(Age, Base, Long, Abs_Diff) %>%
  head(10)

```

	Age	Base	Long	Abs_Diff
1	70	13.00559	13.85579	0.8502058
2	69	13.47172	14.32079	0.8490715
3	71	12.53447	13.38340	0.8489241
4	68	13.93484	14.78183	0.8469923
5	72	12.06494	12.91192	0.8469767
6	67	14.39376	15.23723	0.8434783
7	73	11.59088	12.43392	0.8430365
8	66	14.84918	15.68800	0.8388159
9	74	11.11674	11.95395	0.8372143
10	65	15.29964	16.13241	0.8327728

To fully understand the risk profile, we must also look at the relative impact of the shock. We calculate the ratio of the shocked liability to the baseline to see which age groups experience the largest percentage increase.

```

> df_mort_shocks %>%
  mutate(
    Abs_Diff = Long - Base,
    Ratio     = Long / Base
  ) %>%
  arrange(desc(Ratio)) %>%
  select(Age, Base, Long, Abs_Diff, Ratio) %>%
  head(10)

```

	Age	Base	Long	Abs_Diff	Ratio
1	88	5.142345	5.685293	0.5429481	1.105584
2	87	5.484937	6.061809	0.5768711	1.105174
3	89	4.821826	5.328781	0.5069551	1.105138
4	86	5.846589	6.455410	0.6088203	1.104133
5	90	4.526371	4.995195	0.4688240	1.103576
6	85	6.227048	6.866217	0.6391690	1.102644
7	84	6.625874	7.293580	0.6677065	1.100773
8	83	7.036868	7.730435	0.6935673	1.098562
9	82	7.458677	8.176300	0.7176227	1.096213
10	81	7.888815	8.628485	0.7396698	1.093762

While the absolute financial gap peaks around age 70, the relative percentage increase is highest for the oldest participants (ages 80-90). Because baseline mortality at advanced ages is quite high, a 20% reduction yields a huge increase in their expected remaining lifespan. For example, the liability rises over 10% for a 90-year-old.

Consequently, evaluating both absolute and relative metrics gives that older participants are the most sensitive to longevity shocks. The absolute capital cost hits the retirement-age cohort hardest, while the highest relative percentage jump occurs at advanced ages. Therefore, the Gold Portfolio, which is heavily concentrated in these older demographics holds the highest demographic risk.

Interactive Mortality Shock Simulator

An interactive Mortality Shock Simulator is developed to visualize demographic risk assessment. Figure 17 displays the interface, which allows the user to dynamically adjust the mortality multiplier and observe the resulting impact on cohort life expectancy and expected present value across all ages.

Dynamic Mortality Shock Simulator

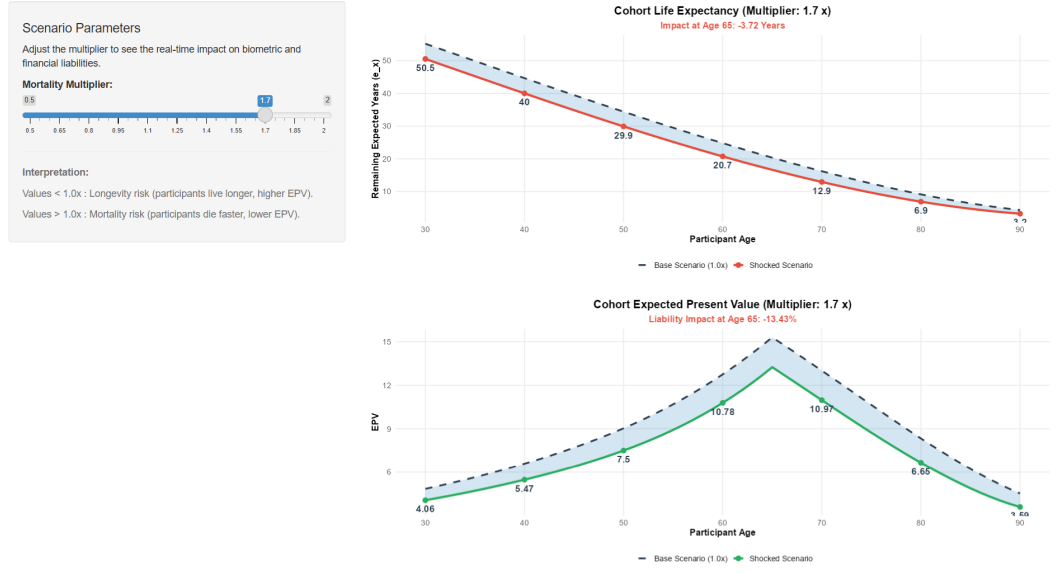


Figure 17: Interactive Mortality Shock Simulator.

2.4 Financial Shocks: Interest Rate Sensitivity

Because pension liabilities are spread over years, their present value is sensitive to discount rates. To assess this financial risk, we evaluate the liability across different economic environments using the EIOPA Risk-Free Rate term structures released between 2016-2022 and 2026. Figure 18 illustrates the evolution of these yield curves.

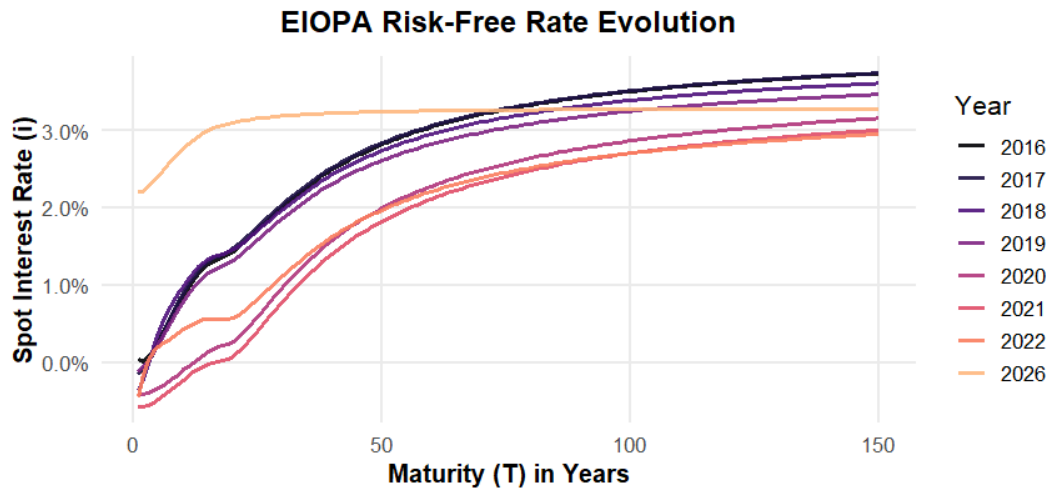


Figure 18: EIOPA Risk-Free Rate Term Structure Evolution.

The macroeconomic environment shifted over the last 10 years. The years from 2017 to 2022 represent a period of spot rates turning negative at the short end of the curve while 2026 curve shows a higher-yield environment.

To quantify the impact of these shifts on the pension fund, Figure 19 evaluates the EPV of the baseline cohort liability under each yield curve.

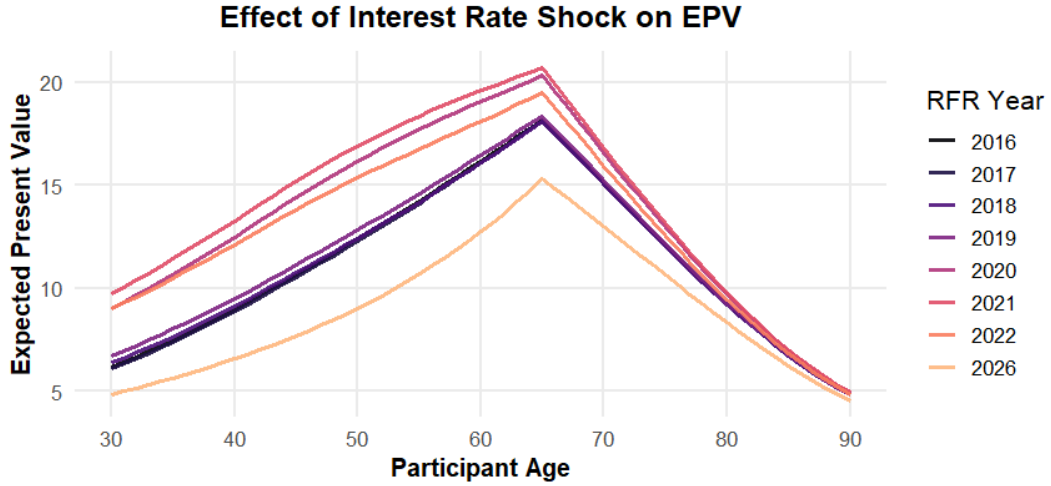


Figure 19: The impact of changing EIOPA discount curves on the EPV.

As expected, there is an inverse relationship between the interest rate and the liability valuation. The low-rate environments produce the highest liability valuations. This represents a financial stress scenario for the company. On the other hand, the higher rates of 2026 heavily discount future cash flows, resulting in suppression of the present value of the obligations.

The vertical gap between the curves shows that the interest rate sensitivity is highly dependent on age. For older ages, the time of cash flows are not late, so the time-to-maturity (T) is short. This has minimal effect on the discount factor $(1+i)^{-T}$, thus on the EPV. Consequently, EPV curves cluster on the right side of the chart. However, for a 30-year-old participant, cash flows are deferred for a long time. So, the discount factor compounds the interest rate change over decades. Consequently, a drop in interest rates increases the present value of their deferred annuity.

Lastly, the shape of the EPV curve depends on the yield curve. It depends on the rate level and convergence speed of curve to its ultimate rate level. The 2026 curve has high rate and also reaches the ultimate forward rate rapidly. This heavily discounts long-deferred cash flows. So, the result is a convex EPV profile. On the other hand, it takes longer time 2021 curve to converge and this diminishes the effect of discounting., So we have a concave (linear-like) EPV curve. Younger participants therefore become capital-intensive to reserve.

This establishes a risk difference for the company. While the older Gold Portfolio holds the highest risk against demographic longevity shocks, younger Green Portfolio is more exposed to macroeconomic interest rate shocks.

2.5 Stochastic Interest Rates: Vasicek Model

We now simulate future interest rate paths to perform a financial stress test with the Vasicek model where we generates stochastic interest rate paths [5]. We determine to four parameters for the evolution of the curves: the initial short rate (r_0), the long-term mean level (θ), the speed of mean reversion (k) and the market volatility (β). Figure 20 displays five simulated yield curves.

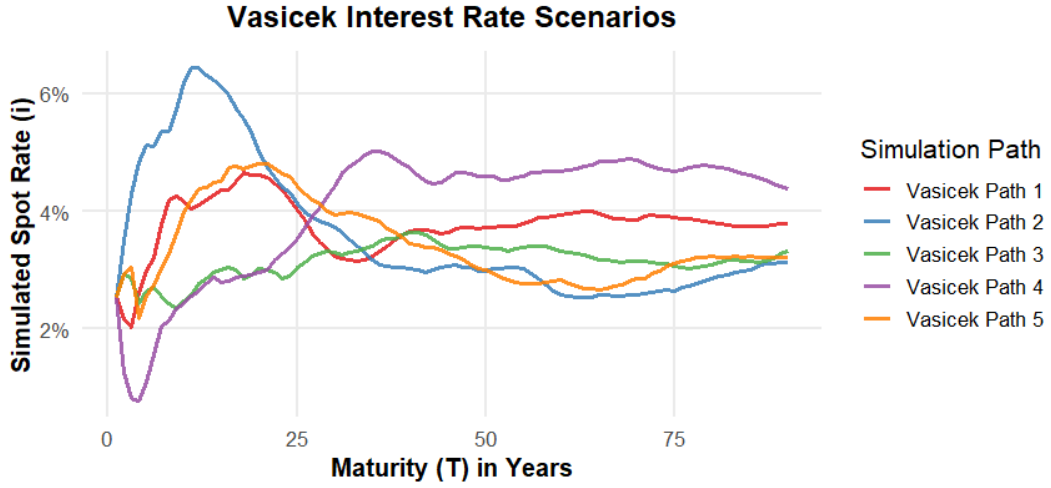


Figure 20: Simulated spot rate curves using the Vasicek model ($n = 5, T = 100, r_0 = 0.025, \theta = 0.035, k = 0.15, \beta = 0.015$).

The random paths visualized here were generated earlier in an unseeded global environment. Reproduction scripts are seeded to guarantee identical structural volatility, however simulated paths will be different.

We focus on Path 2 and Path 4 to highlight the differences and evaluate their specific effects on liability valuation. Path 2 represents a short-term high-rate regime where the spot rate spikes above 6% around maturity year 10 then reverts downward toward the long-term mean. On the other hand, Path 4 represents a short-term suppressed regime where the spot rate drops below 1% in the first five years before increasing to stabilize above the mean for higher durations. These distinct curve shapes impact the liability valuation, as we illustrate in the EPV distribution across ages in Figure 21.

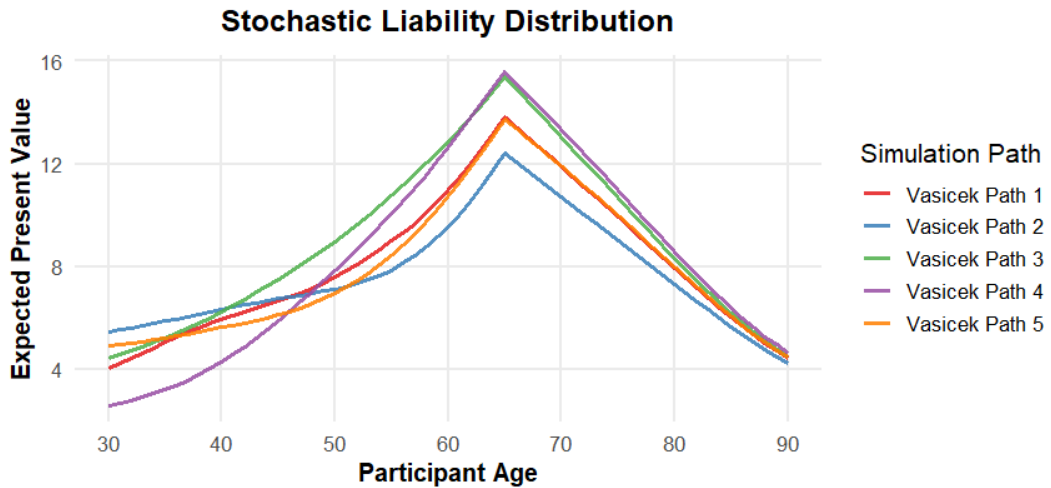


Figure 21: Cohort EPV evaluated under the simulated Vasicek interest rate scenarios.

The financial impact depends on cash flow timing. For a 60-year-old participant whose pension payments begin in five years, Path 4 provides extremely low rates during this near-term payout window. This less discounting rises their EPV to the one of the highest level among all scenarios. Conversely, Path 2 features very high rates during this same window. So, the near-term cash flows become heavily discounted and this drives the EPV to the lowest level.

Conversely, a 30-year-old participant has a different situation because their cash flows are deferred for 35 years. Path 4 features high interest rates at these long maturities. This discounts the distant

cash flows severely and generates the lowest EPV. On the other hand, Path 2 features lower rates at these same maturities and this generates a relatively higher EPV for the young participant.

These stochastic simulations show distinct portfolio vulnerabilities based on age composition. Older portfolios face severe valuation risks from short-term interest rate dips, making the Gold Portfolio highly exposed to regimes like Path 4. Younger portfolios face risks primarily from long-term interest rate declines, meaning the Green Portfolio requires hedging against these environments.

3 Stochastic Mortality Scenarios

3.1 Simulating Future Mortality

While deterministic shocks provide a standard baseline for stress testing, they do not assign explicit probability metrics to extreme demographic events. To quantify these risks, we apply stochastic modeling. We simulate 1,000 future paths for the Lee-Carter mortality trend parameter (k_t). This simulation relies on the historical time-series properties of k_t . The model projects future values by assuming a random walk with drift. It generates distinct paths by adding normally distributed random errors at each annual time step.

Figure 22 displays this stochastic evolution. As the projection extends further into the future, the simulated paths fan out. This represents the compounding uncertainty of mortality improvements over time visually. Each individual path constitutes a future mortality scenario. To ensure every simulated life table spans the entire portfolio horizon, we apply the Kannisto method to close the rates up to age 120.

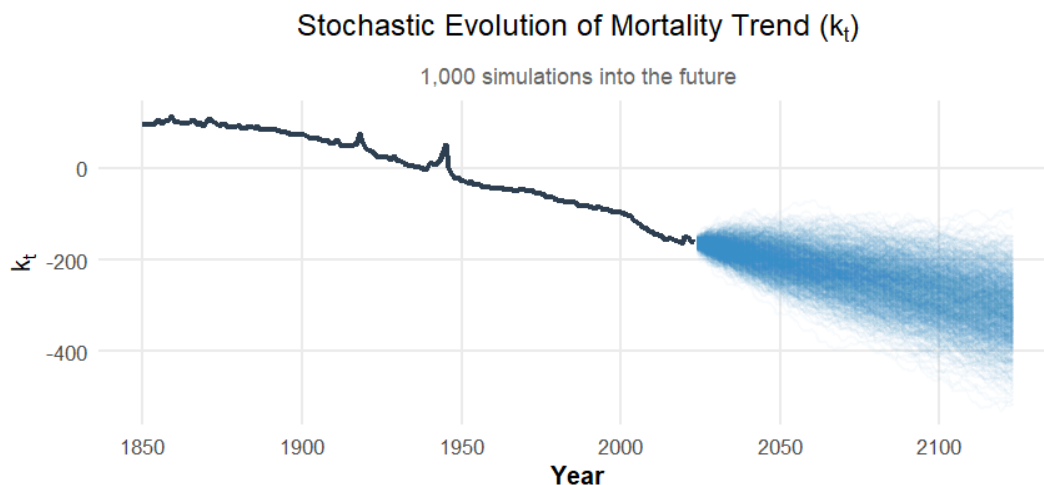


Figure 22: Stochastic evolution of the mortality trend (k_t) across 1,000 simulations.

3.2 Life Expectancy Distributions

Using these simulated mortality trend, we calculate the life expectancy under each scenario for all portfolio ages. Figure 23 illustrates the probability density of these expectations.

The distribution shapes differ structurally by age. Younger cohorts exhibit much wider distributions as the long projection horizons allow the stochastic variance in the k_t parameter to compound. Conversely, older cohorts have a shorter remaining lifespan. This limits the impact of future mortality uncertainty and results in narrow and concentrated density peaks.

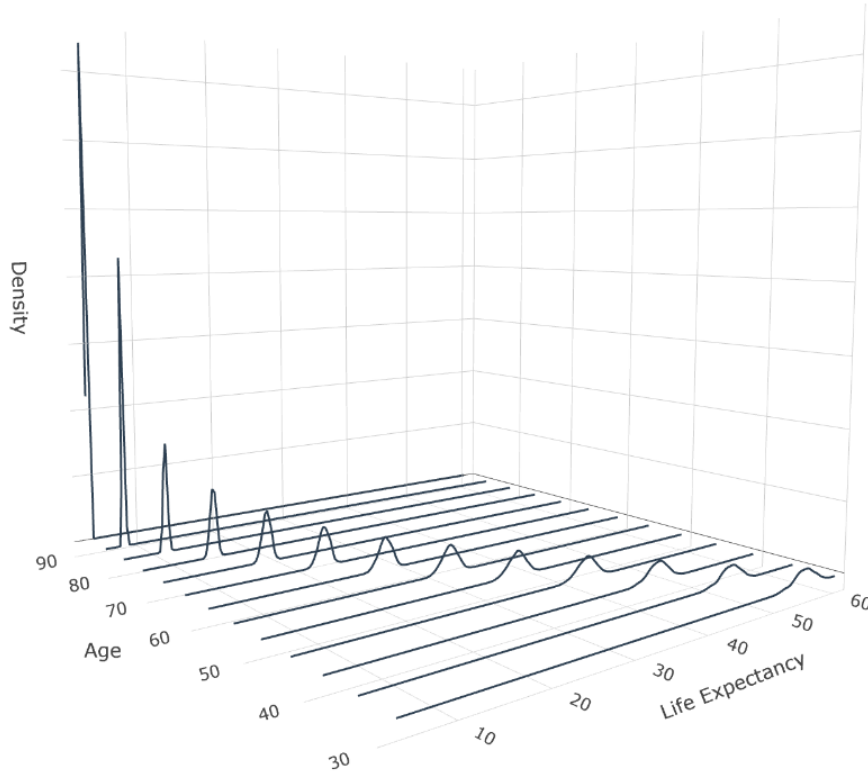


Figure 23: Life Expectancy distributions across portfolio ages.
An interactive, 3D web version of this plot can be found [here](#).

To compare stress scenarios to our stochastic simulations, we isolate the 50-year-old cohort. Figure 24 plots the simulated life expectancy distribution against the deterministic shocks calculated previously. The distribution roughly follows a normal bell curve centered around the base cohort estimate.

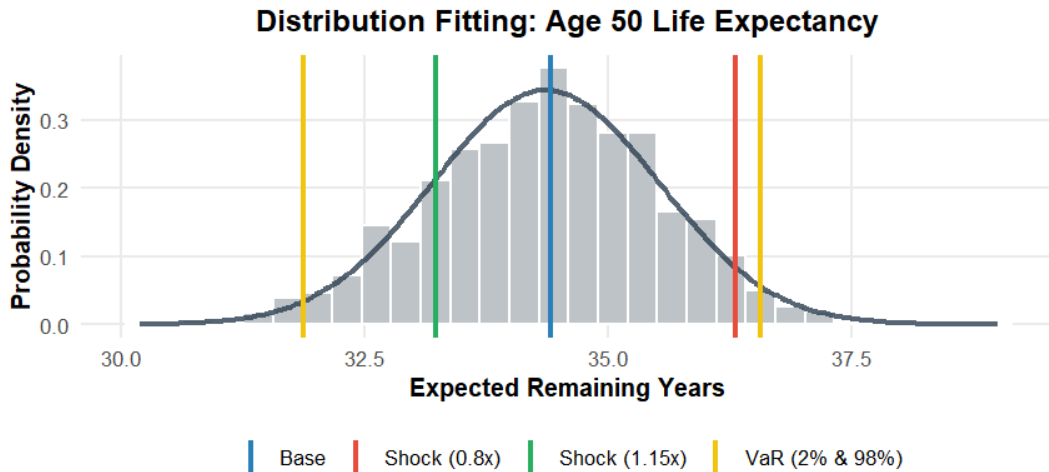


Figure 24: Distribution of Life Expectancy for a 50-year-old against deterministic shocks.

We define the 2% and 98% Value at Risk (VaR) boundaries to represent severe demographic events. The yellow vertical lines indicate these stochastic limits while red and green lines represent the static Solvency II shocks.

We observe that the 98% VaR aligns almost perfectly with the 0.8x longevity shock. So, deterministic multiplier acts as a proxy for longevity risk.

However, the opposite tail reveals a different structure. The 1.15x mortality shock falls short of the 2% VaR boundary. It is even closer to the base mean and it fails to capture the full extent of the stochastic downside risk. This indicates an asymmetric relationship between the two methodologies. The static framework captures extreme life extensions for this cohort but it underestimates the severity of early mortality events modeled by simulations.

3.3 Stochastic Liability Valuation

To translate these demographic distributions into financial liabilities, we calculate the EPV of an annuity for every simulated path with the base 2026 EIOPA yield curve in Figure 25 across ages. We see a similar behavior in this figure where the younger cohorts exhibit much wider EPV distributions as long deferral periods compound the stochastic mortality variance, while older cohorts show narrow and concentrated peaks.

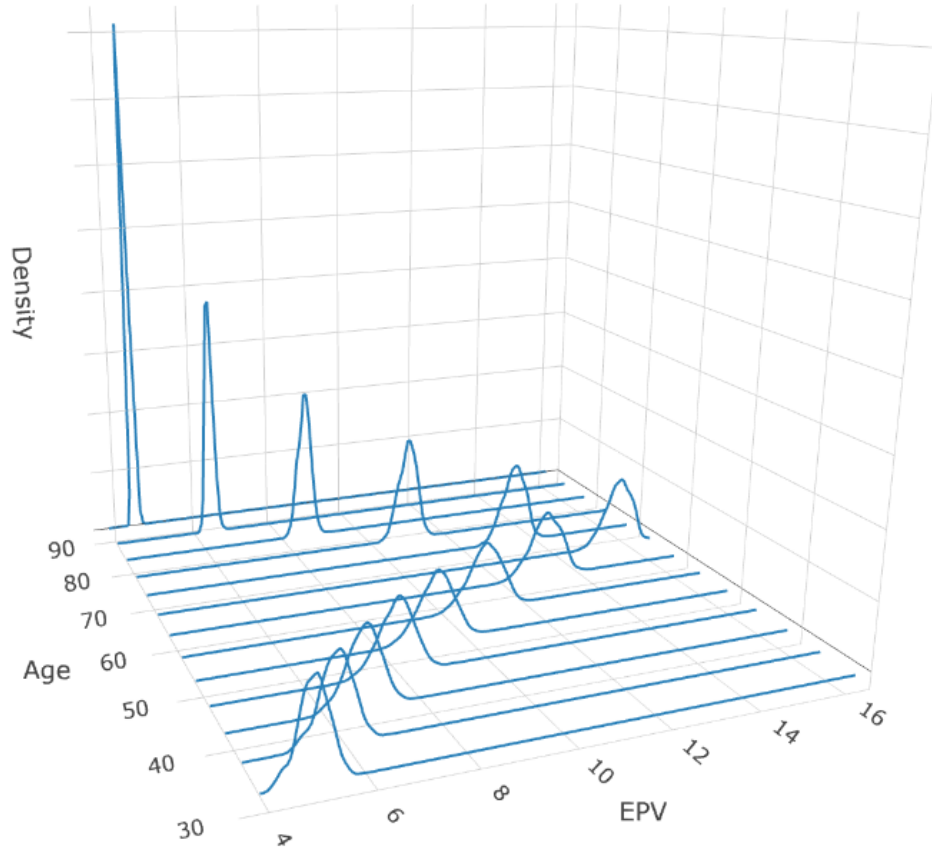


Figure 25: Expected Present Value distributions across portfolio ages.

Note: An interactive, 3D web version of this distribution can be found [here](#).

Figure 26 isolates the financial distribution for the 50-year-old cohort. We compare the stochastic VaR boundaries against the financial liabilities calculated under the deterministic shocks.

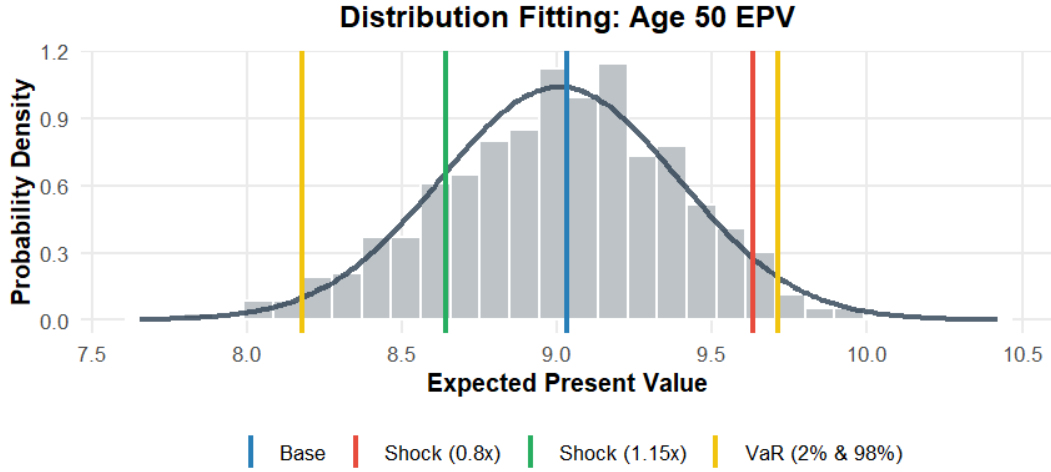


Figure 26: Distribution fitting of Expected Present Value for a 50-year-old.

The right tail of this distribution represents longevity risk and extended lifespans increases the present value of the annuity as it can be observed from the figure. The 98% financial VaR aligns inline with the liability calculated under the 0.8x longevity shock and it prices the extreme longevity risk for 50 year-old as a good proxy.

Conversely, the left tail shows an early mortality where the expected payout duration decreases. In this case, the liability under the 1.15x mortality shock fails to reach the 2% VaR boundary, so it underestimates the financial impact of extreme downside mortality scenarios.

3.4 Age-Dependent Capital Adequacy

While deterministic shocks provide an approximation for specific cohorts like the 50-year-old above, its validity across different ages varies. Figure 27 visualizes the percentage gap between the static multipliers and the stochastic VaR boundaries across all portfolio ages.

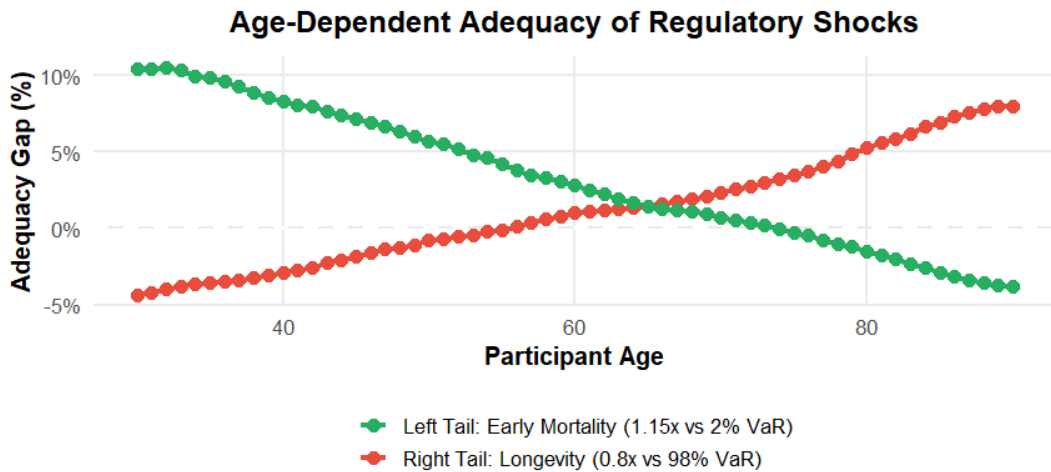


Figure 27: Age-Dependent Adequacy of Regulatory Shocks compared to Stochastic VaR.

The empirical data shows that the capitalization drifts depending on the age. We, focus first on the longevity risk which is represented by the red curve (line). For a 30-year-old, the 0.8x shock under-reserves capital and it falls roughly 4.5% short of the 98% stochastic VaR originating because of similar reasons as above such as young cohorts have years for medical improvements etc. This

exceeds the linear impact of a static multiplier. Conversely, for a 90-year-old, the same shock over-reserves the liability around 8.0% because it assumes an unrealistic acceleration of old-age survival.

On the other hand, the green curve shows the inverse relation. It overestimates sufficient amount of capital by 10% more than 2% VaR value for younger cohorts. However, it fails to reach the 2% VaR boundary for participants over age 75. This underestimates the financial impact of mortality shocks in old age.

Interestingly, the two curves intersect the x axis at different ages. So, there is no cohort where both deterministic shocks provide accurate capital metrics at the same time. This shows that a single static multiplier is not a necessary tool and it can not reflect the true risk profile of a pension fund. Because of this, the total capital requirement depends entirely on the fund's specific age distribution. So, it might be harmful to measure the solvency situation of an institution just by looking at the isolated age groups and portfolios. Therefore, we must combine the cash flows of the entire population using our simulated mortality paths.

4 Portfolio Valuation

4.1 Individual Cashflow Projections

To value the aggregate liabilities, we first project the expected cashflows for an individual. We define the expected payout at any future time T as the annuity amount multiplied by the probability of surviving to that time when retiree reaches the age of 65, else 0. Figure 28 visualizes this cashflow evolution, an active 50-year-old participant against a retired 70-year-old participant.

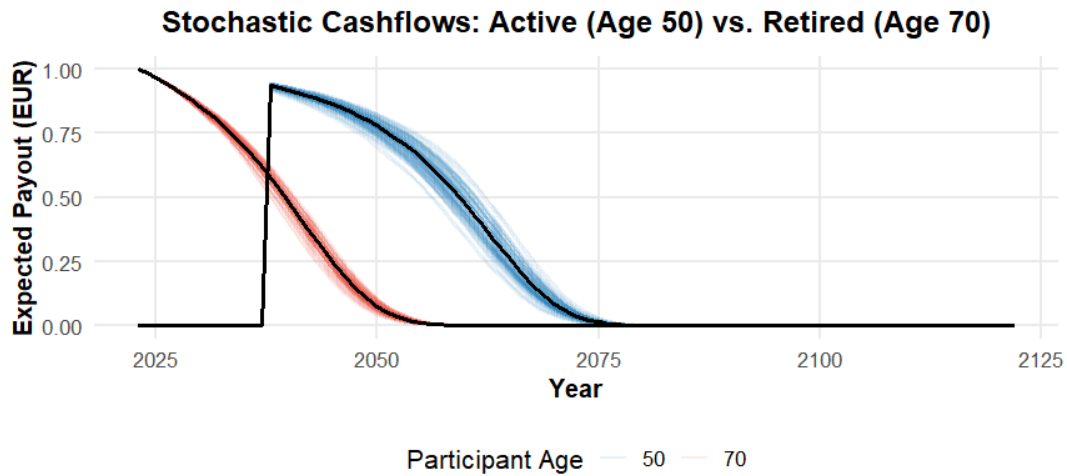


Figure 28: Expected cashflow projections deferred and immediate annuities under 100 stochastic mortality paths.

The structure of the pension liability is determined by the participant's age as follows:

The Active Cohort (Age 50): This represents a deferred annuity. The expected payout remains zero during the 15-year deferral period. At age 65, the expected cashflow spikes and reaches its highest value. This is basically the cumulative probability of surviving to retirement while the surrounding stochastic fan illustrates demographic uncertainty. Because a 50-year-old has years for (dis)improvements to compound before reaching old age, dispersion becomes wider.

The Retired Cohort (Age 70): This represents an immediate annuity. The participant is already over age 65, so the cashflow initiates immediately in the valuation year at exactly 1.00 EUR (as the survival probability at $T = 0$ is 100%). From there, the expected payout begins to decay as natural mortality reduces the cohort.

By applying this cashflow mechanism to every age group and multiplying by the respective participant counts, we map capital requirements for the portfolios for next years.

4.2 Aggregate Portfolio Cashflows

By evaluating the individual cashflows alongside the demographic composition of each portfolio, we can project the total capital requirement. Figure 29 illustrates the best estimate of these aggregated cashflows over the next 100 years.

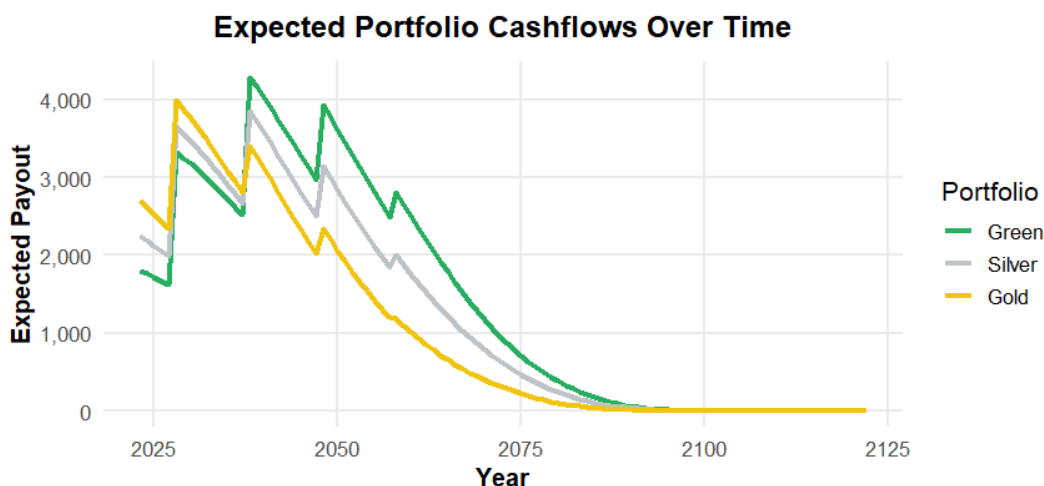


Figure 29: Aggregate expected cashflows for the Green, Silver and Gold portfolios

The sawtooth-like pattern occurs because the portfolio data is grouped into 10-year age cohorts. Each sudden spike represents a younger age cohort reaching the retirement age of 65 and initiating their annuities. For instance, the spike at year 2031 corresponds to the payment of the people who are currently at age 60. Other than that, the graph also shows different risk profiles across the portfolios:

- **The Gold Portfolio (Liquidity Risk):** As gold portfolio represents an aging demographic, its peak cash requirements occur almost immediately. While the total liability finished rapidly with respect to others portfolio, it require liquid, short-term assets to fund the near-term outflows.
- **The Green Portfolio (Duration Risk):** As green portfolio is mostly consisting of younger active members, its cashflow requirements are deferred. While it demands less immediate liquidity compared to others, its liability tail extends most into the future. This creates duration risk and long-term asset matching is important in this case to hedge the portfolio against volatility in interest rates in the future.

4.3 Total Liability Valuation and Uncertainty

To calculate the Present Value (PV) of these liabilities, we discount the aggregate cashflows using 2026 EIOPA curve. However, a deterministic best estimate is not a good example of proper risk management and might result in insufficient for capital reserving. So, to quantify demographic uncertainty, we evaluate the total portfolio liabilities across all 1,000 stochastic mortality paths generated in Phase 3. Figure 30 displays the probability distributions of the total present values. The figure shows that portfolios do not just differ in total cost but they also differ in uncertainty.

- Gold Portfolio displays a narrow and concentrated distribution. Because the payout horizon is relatively short, there is not much time for a change in the mortality rates that affects the total cost. Consequently, its present value is highly predictable with minimal tail risk.

- Green Portfolio shows a widely dispersed and apparent left-skewed distribution. The long projection horizon allows mortality uncertainty to compound significantly over time. This results in high variance, meaning the ultimate liability has a much wider range of potential outcomes compared to older cohorts.
- Silver Portfolio shows moderate dispersion due to its mixed participant ages. It carries more demographic uncertainty than Gold but avoids the extreme variance seen in the younger cohort. So, it lies somewhere in between Gold and Green portfolios in every sense.

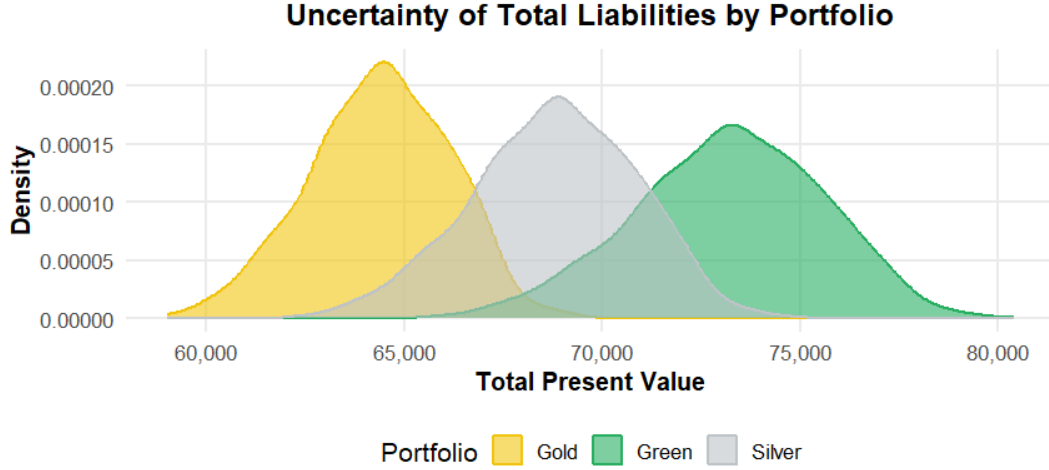


Figure 30: Distribution of Total Present Value for each portfolio across mortality scenarios.

Consequently, while the Green one holds the largest absolute liability, it also carries the highest relative variance. Therefore, the company can't (shouldn't) apply a uniform capital charge across the company. In the below, there is a list of comparison between the Best Estimate and the 98% VaR.

- **Green Portfolio:** Best Estimate of 73,189 EUR; 98% VaR of 77,618 EUR (Ratio: **1.061**).
- **Silver Portfolio:** Best Estimate of 68,778 EUR; 98% VaR of 72,589 EUR (Ratio: **1.055**).
- **Gold Portfolio:** Best Estimate of 64,368 EUR; 98% VaR of 67,605 EUR (Ratio: **1.050**).

The VaR-to-Best-Estimate ratio might be considered as an approximation for the capital intensity of each portfolio. The Green portfolio requires a 6.1% risk margin to cover 1-in-50-year longevity events, whereas the Gold portfolio requires only 5.0%. This 1.1% gap shows that the tail of a younger demographic has a higher impact on capital reserving.

4.4 Sensitivity Analysis: Biometric vs. Financial Shocks

In this section, we evaluate how the total present value reacts to regulatory and financial shocks to assess portfolio risk. We focus on the percentage change relative to the base valuation to compare the portfolios.

4.4.1 Longevity and Mortality Shock

We first apply the deterministic multipliers: a 0.8x factor for a longevity shock and a 1.15x factor for a mortality shock. Figure 31 illustrates the relative impact of these scenarios.

The relative impact of these shocks is almost uniform across portfolios. The longevity shock increases the liability by 6.5% to 6.7% across all portfolios, while the mortality shock decreases the liability by an identical 4.2%.

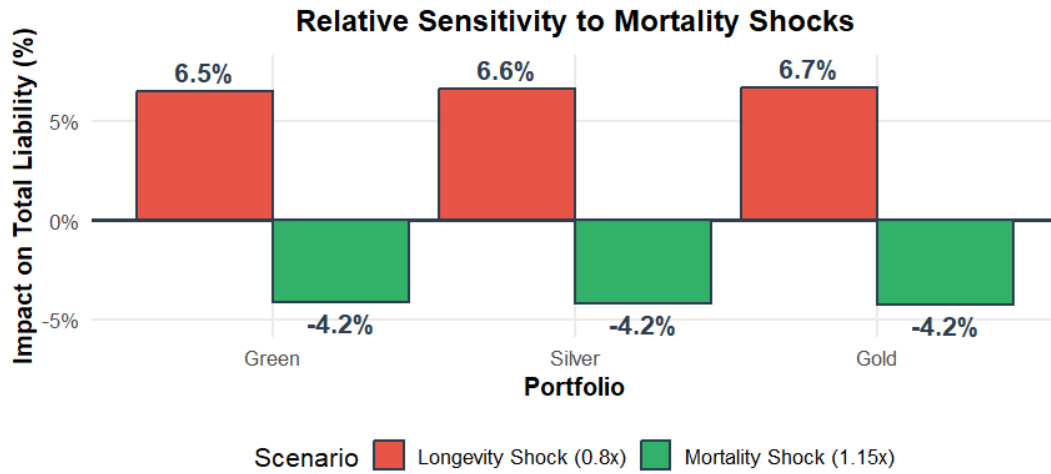


Figure 31: Relative impact of Solvency II biometric shocks on Total Present Value.

Given that younger cohorts have longer future projection time, it might be expected that the Green portfolio shows a much higher percentage sensitivity to longevity. However, this biological exposure is neutralized by the time value of money. For the Gold portfolio, extended life expectancies generate extra cash flows in the near term and these cashflows face minimal interest rate discounting. For the Green portfolio, the extra survival years occur 50 to 60 years in the future. The heavy discounting applied to these distant cash flows cancels out the compounding mortality effect. If interest rates were uniformly zero or very close to zero, the discount factor would remain around 1.0 for all maturities and this would eliminate this neutralization effect.

4.4.2 Interest Rate Sensitivity

Next, we evaluate financial sensitivity by applying parallel shifts of +100 and -100 bps to the EIOPA curve. Figure 32 displays the results.

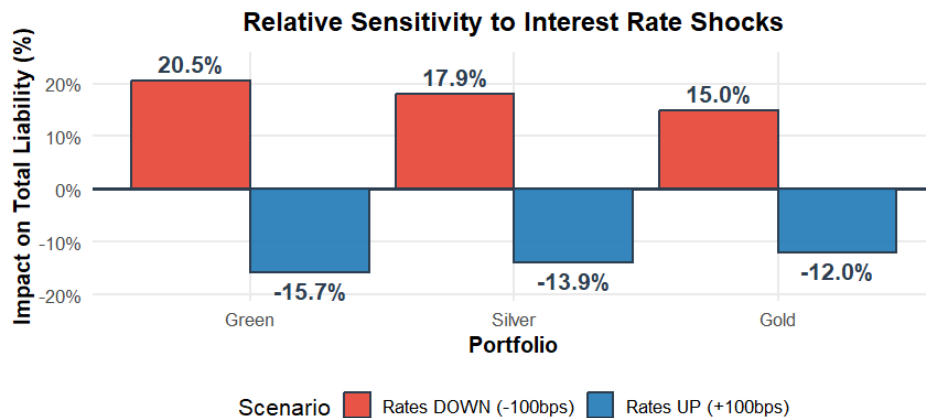


Figure 32: Relative impact of parallel interest rate shifts on Total Present Value.

Unlike the mortality shocks, we have a huge differences between the portfolios. The Green portfolio is highly sensitive to interest rate movements, with its liability swinging between +20.5% and -15.7%. In contrast, the Gold portfolio shows lower sensitivity (+15.0% to -12.0%).

This variance can be explained by duration risk. An interest rate shock changes the discount factor $v^T = (1 + i)^{-T}$ and as the Green portfolio consists of deferred annuities, its cash flows occur much later (larger T). So, a +1% shift in the interest rate compounds exponentially over these longer

time horizons. In conclusion, while the longevity shocks impact all portfolios more or less equally in relative terms, the Green portfolio is exposed to interest rate volatility the highest.

5 Conclusion and Recommendations

In this report we evaluated the liabilities of the company across different portfolios. By transitioning from static period life tables to dynamic cohort projections and applying stochastic simulations we identified the risk factors. To ensure an adequate capital reserving, the followings are recommended:

1. Cohort-Based Reserving

Relying on period life expectancy structurally underestimates future liabilities. Because the Green portfolio consists primarily of young active members, ignoring future mortality improvements results in a capital shortfall. So, the company needs to adopt cohort-based valuation for reserving.

2. Duration Risk in the Green Portfolio

The Green portfolio represents the largest long-term liability. Sensitivity analysis shows that its deferred cash flows make it highly vulnerable to interest rate fluctuations. A 100 bps drop in interest rates inflates its liability by over 20%. Thus, ISSurance must implement an asset matching strategy like investing in long-term bonds to hedge against in case yield curves fall.

3. Liquidity Risk in the Gold Portfolio

While Gold portfolio is relatively protected from interest rate shifts, it requires immediate cash outflows as the participants are mostly elderly people. So, the investment strategy for this portfolio must be in the assets that are liquid and short-term assets to fund annuity payments.

4. Dynamic Risk Margins

The stochastic simulations reveal that uncertainty is not uniformly distributed. The Green portfolio exhibits a wider dispersion of tail risk compared to the Gold portfolio. So, company should apply age-dependent risk margins that takes compounding mortality uncertainty of different ages into account.

A Proof of Peak Expected Present Value at Retirement Age

For a participant aged $x < 65$, the expected present value of the deferred annuity is defined as $EPV(x) = v^{65-x} \cdot {}_{65-x}p_x \cdot \ddot{a}_{65}$. The relationship between consecutive ages can be evaluated using the backward recursion formula: $EPV(x) = v \cdot p_x \cdot EPV(x+1)$. Assuming a strictly positive interest rate, which is the case for the Netherlands, the one-year financial discount factor $v = (1+i)^{-1}$ is less than 1. Because the one-year survival probability $p_x < 1$, their product must satisfy $v \cdot p_x < 1$. Substituting this into the recursion yields $EPV(x) < EPV(x+1)$. So, the liability monotonically increases each year during the deferral period.

Conversely, for a participant in the payout phase ($x \geq 65$), the liability simplifies to an immediate whole life annuity: $EPV(x) = \ddot{a}_x = \sum_{t=0}^{\infty} v^t \cdot {}_t p_x$. Due to the mortality dynamics, force of mortality increases with age. This implies that for any duration $t > 0$, the cumulative survival probability of a younger individual is greater than that of an older individual, so we have ${}_t p_x > {}_t p_{x+1}$. Because the financial discount curve v^t applies identically across both summations, the term-by-term comparison dictates that $\sum v^t {}_t p_x > \sum v^t {}_t p_{x+1}$, yielding $EPV(x) > EPV(x+1)$. Because the sequence monotonically increases for $x < 65$ and monotonically decreases for $x \geq 65$, the global maximum of the EPV occurs at $x = 65$.

B Source Code and Reproducibility

For reproducibility, the R codebase is available in the link below.

Github Repository: github.com/keremiis/ALIM-HW

- **Phase 1 (Historical Data & Period Valuation):** HMD data, Lee-Carter model calibration, Kannisto tail extrapolations and period life expectancy calculations.
[\[phase_1.R\]](#)
- **Phase 2 (Cohort & Deterministic Shocks):** Transition to cohort survival probabilities, historical interest rate shocks and Vasicek interest rate scenarios.
[\[phase_2.R\]](#)
- **Phase 3 (Monte Carlo Simulations):** Monte Carlo simulations for the Lee-Carter k_t parameter, 3D density distributions and VaR checks.
[\[phase_3.R\]](#)
- **Phase 4 (Portfolio Valuation):** Expected cashflow generation, aggregate portfolio liabilities, 98% VaR metrics and sensitivity analysis.
[\[phase_4.R\]](#)
- **Interactive Dashboard:** Application for mortality shock simulation.
[\[app.R\]](#)

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